# Ageing Institutions Ossify a Cultural Field: Evidence from Soviet Book Publishing for Children

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**Abstract**

Some cultural fields undergo an ossification process, with an ever larger share of the produced material featuring already established authors. I argue that this can be explained by the ageing of the institutions of cultural production, without assuming that anyone changed their behavior. The core mechanism is the growing backlog: the probability that a producing unit fills a slot with an item from outside its own catalog diminishes with the size of that catalog, so a field that stops admitting new producing units will ossify. I test this mechanism on the case of late-Soviet book publishing for children, using digitized bibliographic data covering 1918–1984. Fitting a model in which a board fills a slot with an author new to it at a probability that falls with the size of its accumulated catalog, I find no evidence that the propensity of the boards to publish authors new to them declined over 1950–1984; if anything it rose. Counterfactual tests show that the exposure of an editorial board to its own growing catalog is both necessary and sufficient to explain the diminishing rate at which boards take on new authors.

Keywords: cultural change, neutral models, institutions, children's literature, book history

## 1 Introduction

Some cultural fields undergo what a casual observer may call an ossification process, with an ever larger share of the produced material featuring the already established authors, familiar universes, and recognizable characters. Ossification as a long-running trend has been discussed recently for movies, TV shows, music, video games, and books (Mastroianni 2022). I argue that this effect can be explained by the ageing of the institutions of cultural production, such as publishers and studios, without assuming that anyone changed their behavior.

The growing backlog of published material as the producing institutions age is the core mechanism for the ossification of the repertoire. In the simplest model the probability that a producing unit fills a slot with an item from outside its own catalog diminishes with the size of the catalog. When a field stops admitting new producing units, ossification will inevitably ensue. This argument introduces the logic of the neutral-null theory of cultural change at the level of the institutional structure of the field.

In this paper, I test the institutional ageing mechanism on the case of late-Soviet book publishing for children. This case provides a well-defined field with an especially legible administrative history thanks to Soviet bibliographic and publishing institutions. The quantitative study is made possible by the available digitized bibliographic data covering 1918–1984 (Maslinsky 2022).

Ossification is clearly visible in these data. By the end of the period, the share of printed copies that went to already well-established authors grew five-fold, while the share of editions allocated to new authors fell. In addition, after 1975 new editorial boards almost ceased to appear in the bibliographic record, while the overall number of active boards did not fall.

I fit a model in which the probability that a board prints an author new to it decreases as the board's own catalog grows. The model separates this effect of catalog size from a field-wide, year-by-year propensity to take on new authors, which I call *recruitment intensity*. There is no evidence that recruitment intensity declined over 1950–1984; if anything it rose. Hence the apparent ossification is explained by the ageing population of editorial boards with their accumulated publishing catalogs. Counterfactual tests show that the exposure of the editorial board to its own growing catalog is both necessary and sufficient as a mechanism for the diminishing rate at which boards take on new authors.

## 2 Related work

At first sight, the relatively higher share of well-established children's authors by the end of the Soviet period can be read as the effect of accumulated individual success. Cumulative advantage is a well-known phenomenon in cultural fields (Merton 1968), and rising concentration is typically taken as evidence about the dispositions of whoever is doing the selecting. In literary studies, cumulative advantage has been explained by accumulated symbolic capital (Bourdieu 1993), and by the existence of specialized institutions that reproduce status and hence participate in the canon formation (Guillory 1993). As was shown in a famous empirical paper, a consumer in a cultural market resolves their uncertainty by relying on the choice of others, and thus success breeds success (Salganik, Dodds, and Watts 2006). But alternative generative mechanisms for the observed frequency distributions are rarely considered (DiPrete and Eirich 2006).

Neutral theory in ecology and evolutionary biology is the tradition that offers such an alternative explanation (Hubbell 2001). Neutral-null models assume that there is no selection pressure, and cumulative advantage results from the

random choice of variants proportional to their frequency. Neutral models have been applied in cultural evolution studies as well, e.g., to decoration on ceramics or baby name selection (Neiman 1995; Bentley, Hahn, and Shennan 2004; Kandler and Crema 2019).

The Hoppe urn (Hoppe 1984) is the sequential form of the Ewens sampling formula of neutral population genetics (Ewens 1972), which also underlies neutral biodiversity theory (Hubbell 2001) and has since travelled widely (Crane 2016). The extensive discussion of neutral models in population genetics and ecology led these fields to a realization that the fit of a neutral model to the frequency distribution of variants in a population is weak evidence against selection, and that temporal dynamics are more informative about the effect of selectively neutral mechanisms such as drift (Leroi et al. 2020). I build the model on the Hoppe urn's formula to test the effects of the selectively neutral force, which is in this case the exposure of the editorial decision-makers to their own growing catalog of authors.

Quantitative work aiming to explain diversity and success in the cultural markets rarely models the producing units directly, such as publishers or studios. For instance, the historical dynamics of diversity and concentration of the US recording industry have been studied at the level of the market (Peterson and Berger 1975; Lopes 1992; Dowd 2004). The success of artists has been studied with models of individual career trajectories (Fraiberger et al. 2018; Janosov, Battiston, and Sinatra 2020). I am not aware of studies where the catalog of the producing unit has been the object of analysis.

Neutral models of culture have so far addressed transmission between individuals and reception by consumers. The level of analysis that is missing is the production units themselves and their demography. Organizational ecology has studied the founding, ageing and mortality of producing organizations and noted the change in innovative behavior with the organization's age (Hannan and Freeman 1984; Sørensen and Stuart 2000). My point is that an ageing population of gatekeepers produces market concentration with no change in disposition at all.

# 3 Data

The data are 71,000+ editions of children's books published in the Soviet Union between 1918 and 1984, one row per edition, carrying the print run (*tirazh*), the year, the publisher, the city, and the author (Maslinsky 2022). The print run is planned supply, fixed in advance by an editorial bureaucracy and printed in the imprint of the book itself. The print run figure is thus the recorded quantity that directly reflects the allocation decision by the editorial board.

The producing unit is the *board*—a publishing house together with its normalized city, which in the Soviet system is an editorial board with its own staff, plan and list. Boards are recovered from the combination of the publisher and city in the imprint, and corrected to take into account mergers and renamings. The central children's house (Detizdat, Detgiz, Detskaya literatura) is treated

as one board whatever city its imprint names, since its imprint changes from "Moscow; Leningrad" to a single city in 1954 with no institutional change. Its Leningrad section, with its own staff and policy, is a separate board until its liquidation in 1937.

For each board $b$ and year $t$ I count the author slots filled, $n_{bt}$ (the distinct authors it printed that year); how many of them went to an author the board had not printed before, $n_{bt}^{\text{entry}}$; and the board's accumulated catalog $M_{bt} = \sum_{s<t} n_{bs}$, its author slots accumulated strictly before $t$. $M$ is cumulated over the board's whole observed history. $M$ is the number of slots, not of distinct authors: printing the same 10 authors in three consecutive years gives $M = 30$. A *newcomer* is an author new to the board, who may already have been printed elsewhere. Three filters then apply (Table 1), leaving 3,933 board-years on 566 boards, covering 40,790 author slots.

| Panel | board-years | boards | slots |
|---|---|---|---|
| Board-years, 1940–84 | 4,845 | 1,030 | 42,409 |
| + vintage $\geq 1$ | 4,042 | 616 | 40,955 |
| + $M > 0$ (model panel) | 3,933 | 566 | 40,790 |

Table 1: Construction of the analysis panel. Pre-1940 years are dropped because data are left-truncated at 1918 and that makes returning authors look new, inflating entry counts. A board's first year (vintage 0) is all entry by construction and so carries no information about recruitment. $M > 0$ is required because the exposure term is undefined at $\log 0$; this drops 109 rows, one per board whose first *authored* year lags its first edition of any kind

The idea that the field of children's book publishing ossified in the late-Soviet period rests on two trends apparent in the data. First, at the author level the premium on having published before grew with time. The share of printed copies allocated to authors with more than ten prior editions runs from 0.14 in 1925 to 0.78 in 1984, while the share of author slots at a publishing board going to someone that board had never printed before falls from 0.63 in 1940 to 0.44 in 1984. Second, at the field level, the producing population ages and stops turning over. A median of 41 new editorial boards appears in the bibliography per year between 1923 and 1939, against 2–8 per year after 1975, while 99–133 boards went on printing each year (Figure 1).

# 4 Models

## 4.1 The exposure model

The core model for testing the neutral-null hypothesis is the Hoppe urn (Hoppe 1984), which is equivalent to the Chinese restaurant process (Crane 2016). An urn holds a distinguished "mutator" ball of mass $\theta$ alongside $M$ balls standing for the items already drawn. Any ball drawn is returned to the urn with an

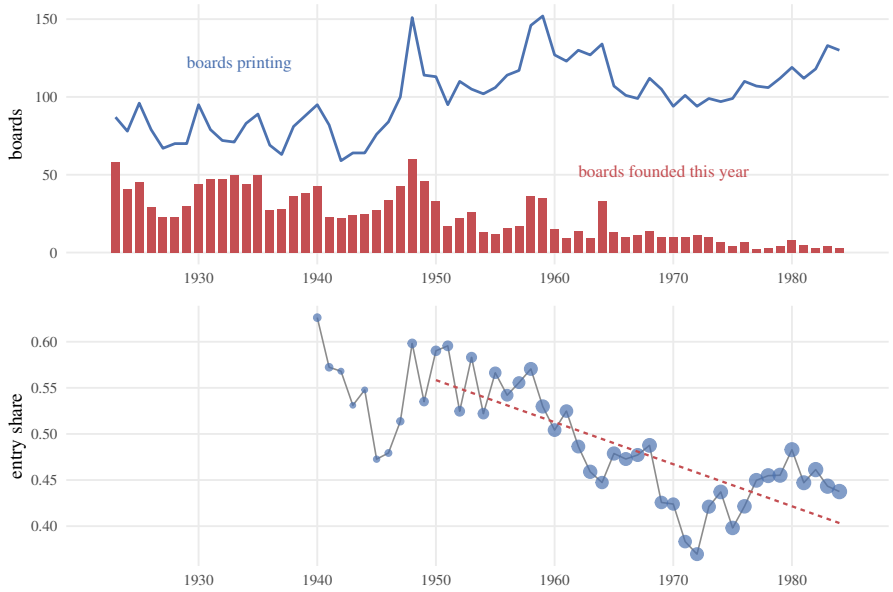



Figure 1: The field stops admitting new producing units (top) and the share of slots going to authors new to the board falls (bottom). Top: boards founded in each year (bars) against the standing population of boards printing (line). Bottom: author slots new to the board as a share of all slots filled, on the model panel; point size is the number of slots, and the dashed line is the 1950–84 linear fit

additional ball of the same kind. If a draw takes the mutator, a ball of a type not seen before is introduced to the urn, with probability

$$p = \frac{\theta}{\theta + M}. \tag{1}$$

Read as a model of a producing unit, $\theta$ is recruitment intensity, i.e. propensity to fetch an author from outside the catalog. $M$ is the number of author slots the board has already filled. Equation 1 then says that the entry probability falls in $M$ with $\theta$ held fixed. This means that an ageing unit admits proportionally fewer newcomers without having become any less inclined to admit them.

In the Hoppe urn the exponent on $M$ is fixed at one. I add flexibility to the model by letting the exponent $\gamma$ vary freely, and allow each board its own level and each year its own effect.

$$\begin{aligned} n_{bt}^{\text{entry}} &\sim \text{Binomial}(n_{bt},\, p_{bt}), \\ \text{logit}\, p_{bt} &= \underbrace{\alpha + \tau_t}_{\log \theta_t} + u_b - \gamma \log M_{bt}. \end{aligned} \tag{2}$$

With $\gamma = 1$ equation 2 is a restatement of equation 1, since $\text{logit}\{\theta/(\theta + M)\} = \log\theta - \log M$. In the parameter space, $\gamma$ measures how far the strength of the exposure force in the publishing field departs from the idealized model of the Hoppe urn. Here $\tau_t$ is a first-order random walk over years with $\tau_1 = 0$, $u_b \sim \text{Normal}(0, \sigma_u)$ is a board effect, and $\gamma$ carries a weakly informative $\text{Normal}(1, 0.3)$ prior centred on the theoretical value. The model is fitted by Hamiltonian Monte Carlo in Stan. The full details of the model specification along with the fitted objects, the simulation experiment data, and various checks are available in the accompanying replication dataset (Maslinsky 2026).

The model takes as given which boards are active each year and how many authors each printed, $n_{bt}$. It explains only how these slots are split between newcomers and authors already in the board's catalog. The year-to-year change is defined by the growth of the catalog size $M$ for all printing boards and by the smoothed history of field-wide recruitment intensity (the random-walk prior on $\tau_t$).

### 4.2 The estimand

The quantity of interest is the linear trend in $\log\theta_t$ over 1950–1984, computed as the ordinary least-squares slope of $\tau$ on year *inside each posterior draw*, so that uncertainty in $\gamma$ propagates into it automatically and no prior asserts a sign. A zero or positive trend means that exposure alone accounts for the observed decline in the newcomer share; a negative trend means that a behavioral change in recruitment is required to explain it.

## 5 Results

### 5.1 Recruitment intensity does not decline

The trend in $\log\theta_t$ over 1950–1984 is +0.0078 per year, on a 90% posterior interval of $[+0.0044, +0.0111]$, with $P(\text{trend} < 0) = 0.00$. It is small and credibly positive, and stays positive when the window starts in 1955 (+0.0056) or 1960 (+0.0064). Figure 2 visualizes this result by showing that recruitment intensity did not fall over exactly the period in which the entry share fell from 0.59 to 0.44. It holds near its 1950 level through 1954 and rises during the Thaw.

The data drive the exposure exponent $\gamma$ into the sub-linear range of roughly 0.54–0.72, but the exact value is not identified well enough. The problem is that the fitted model gives an estimate of 0.54 while a fixed-effects GLM on the same likelihood gives 0.72. The $\gamma = 1$ assumed by the Hoppe urn model is rejected both ways. $\gamma$ is not prior-driven: refitting under a deliberately wide $\text{Normal}(1, 1)$ moves it from 0.537 to 0.535, and the estimand from +0.0078 to +0.0076. The magnitude of the trend depends on $\gamma$, but its sign does not: refitting the year effects with $\gamma$ fixed at values from 0.4 to 1.5, the trend is positive for any $\gamma$ above about 0.45, and reaches +0.0198 at the GLM's estimate.

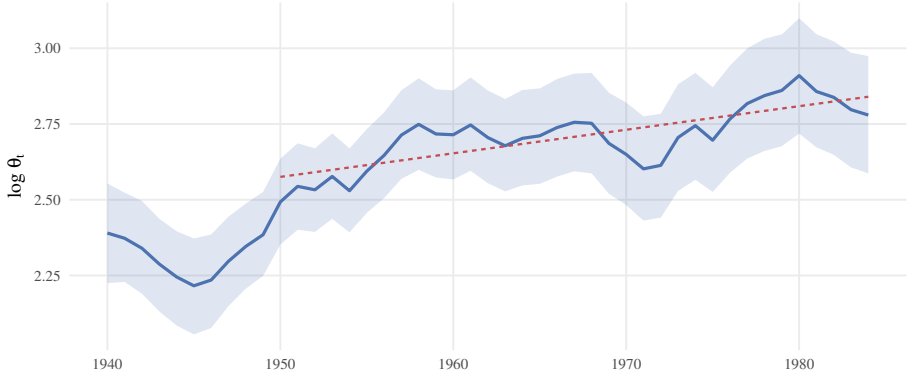



Figure 2: Recruitment intensity, the part of behavior that exposure cannot explain. Posterior mean and 90% band for $\log \theta_t$. The dashed line is the linear trend in $\log \theta_t$ over 1950–1984 (the estimand, Section 4.2), at its posterior mean of +0.0078 per year. Compare the falling series in the lower panel of Figure 1, over the same years and the same boards

## 5.2 Would a decline have been detected?

A finding of no decline is informative only if a real decline would have been found. I checked this on simulated datasets that keep the real boards, years and slot counts $n_{bt}$, and hence the real catalogs $M_{bt}$, and redraw only the number of newcomers, slot by slot, from the Hoppe urn (equation 1). The urn's $\theta_t$ is set year by year to reproduce the observed entries, and a linear decline is then added to $\log \theta_t$ from 1950 on. The model is fitted to eight panels in each of four conditions: the urn's own trend (+0.008 per year), no trend, and declines of −0.01 and −0.02 per year.

Both declines are detected in 8 of 8 datasets (90% interval entirely below zero), and no decline is detected in any of the 16 datasets without one. The estimated trend follows the imposed trend closely (regression slope 1.05, $R^2 = 0.97$), so any decline steeper than about −0.005 per year would be detected. The real estimate, +0.008, is far from it. Because the simulation used exponent 1 and no board effects, unlike the fitted model, every estimate is shifted up by about +0.002 per year. The shift is the same in every condition, so it does not affect whether a decline is detected, and it is too small to account for the real estimate of +0.008. On datasets simulated from the fitted model itself, the estimate runs low by 0.004 per year and its intervals are too narrow, so the real trend's sign holds but its size is uncertain.

## 5.3 The counterfactual

Because $M$ counts slots, recruitment does not affect later exposure: both a newcomer to the board and a returning author add 1 to $M$. $\theta$ and $M$ can

therefore be changed independently. Each counterfactual condition below keeps the real boards, slot counts and board effects $u_b$ and fixes either $\theta$ or $M$. The resulting field-wide entry shares are computed from the model's predictions in the same way as the observed share.

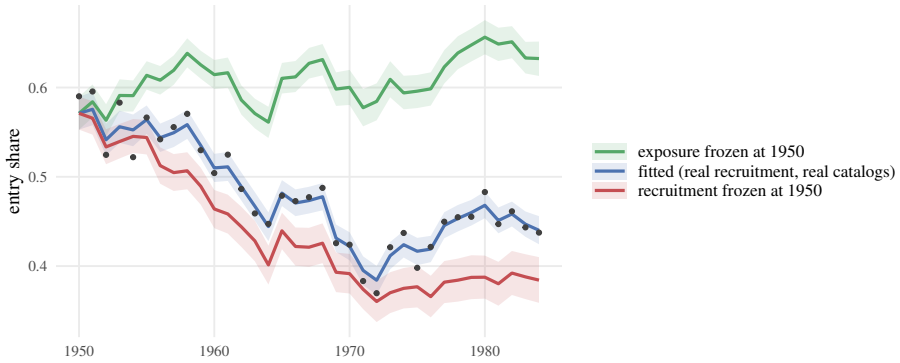



Figure 3: Exposure is sufficient, and it is necessary. Entry share implied by the fitted model under three arms, with 90% posterior bands; grey points are the observed series. All three arms coincide at the 1950 anchor by construction

**Sufficiency.** Freeze the field-wide recruitment intensity $\theta_t$ at its 1950 value and keep the real catalogs: the entry share still falls to 0.384 $[0.359, 0.410]$ by 1984, against a fitted 0.440 and an observed 0.438. That is 144% of the fitted decline, on a 90% interval of $[117\%, 178\%]$. Exposure alone would have moved the share of newcomers lower than in the real data, and the rise in $\theta_t$ offset part of it. If recruitment intensity is fixed at the level of any year between 1953 and 1960, the entry share still falls by 114–136% of the fitted decline, although the fall being explained shrinks, especially after 1957.

**Necessity.** Now let $\theta_t$ take its estimated values but hold catalogs at their 1950 sizes. Only 78 of the 459 boards printing after 1950 existed in 1950, so each year's boards are ranked by catalog size and given the 1950 catalog ($M$) at the same rank: the median board of 1984 gets the median 1950 catalog. The decline not only stops, but reverses, ending at 0.632 $[0.613, 0.652]$—above the 1950 baseline of 0.571.

The reversal comes from recruitment intensity itself. Between 1950 and 1984 the year effect rises by 0.29 on the log-odds scale, while the average board effect among the boards printing each year barely moves ($-0.05$), so the mix of boards printing does not explain it. Once exposure stops pulling the entry share down, the rise in $\theta_t$ becomes visible.

# 6 Discussion

Late-Soviet book publishing for children is a clear case of the ossification of a cultural field. The share of slots going to authors new to the board fell by a

quarter between 1950 and 1984 while recruitment intensity did not fall. Freezing recruitment intensity at its 1950 value still reproduces the fall. So this field-level change can be explained without assuming any change in the behavior of the gatekeepers. What did change was the age structure of the producing population. Then the assumption required for the explanation is the value of the backlog catalog for the established editorial boards, not their propensity to recruit.

There are three important limitations to the model fitted in this study. First, $\theta_t$ absorbs everything that changes besides the exposure — shifts in the author pool, in the general scale of production, in the completeness of the catalog. A rising $\theta_t$ is therefore not evidence of a more open editorial policy: a growing pool of authors would produce it too. It licenses only the claim that the fall in entry requires no behavioral closure. Second, the exposure exponent is weakly identified, and the pooled (Stan) and unpooled (GLM) estimates of $\gamma$ disagree by 0.18 on the real panel against 0.06 on panels simulated from the fitted model. This can be a signature of model misspecification, most plausibly the heterogeneity in the $M$-slope among editorial boards. A board-varying $M$-slope is the specification to try next. Third, entry is measured relative to the board. About half of the entries over 1950–1984 are authors new to the system as a whole (first appearance after 1918), and their share of slots fell from 0.25 to 0.21. That is between a quarter and two-thirds of the fall in the entry share, depending on whether endpoints or linear trends are compared. The exposure model accounts for the intake of the boards, not for the intake of the field as a whole.

The Soviet case is not a typical case of a cultural market: print runs were planned, so reader preferences exerted little direct pressure on the boards' decisions, and institutional barriers to the creation of new boards were structured by state policy rather than by the market. The exposure model itself makes no reference to readers, and commercial publishers also build on their back-lists. But in a free market reader demand may act as a stronger selection force, so field-level closure need not follow. Whether the result holds in a free market thus remains open. Testing this requires comparable catalog-level data from a market field.

The Soviet case is a proof of concept that illustrates a more general idea. Wherever producing units accumulate a catalog and the population of units ages without replacement, the observed entry share is confounded by exposure. To make any judgement about the behavioral change of the institutional agents, the directly observed decline in the share of newcomers is not enough—one needs to account for exposure first. This makes institutional ageing a relevant neutral-null model, similar to a random-copying null already familiar in the study of cultural frequency data (Bentley, Hahn, and Shennan 2004; Kandler and Crema 2019). It is the neutral model to check against before any claim about disposition change is made. Read that way, rising concentration on top performers in a cultural field is not yet evidence about what gatekeepers prefer. It may be evidence about how old they are.

## References


Bentley, R. Alexander, Matthew W. Hahn, and Stephen J. Shennan (2004). "Random Drift and Culture Change". In: *Proceedings of the Royal Society of London. Series B: Biological Sciences* 271.1547, pp. 1443–1450. DOI: 10.1098/rspb.2004.2746.

Bourdieu, Pierre (1993). *The Field of Cultural Production: Essays on Art and Literature*. Ed. by Randal Johnson. New York: Columbia University Press.

Crane, Harry (2016). "The Ubiquitous Ewens Sampling Formula". In: *Statistical Science* 31.1, pp. 1–19. DOI: 10.1214/15-STS529.

DiPrete, Thomas A. and Gregory M. Eirich (2006). "Cumulative Advantage as a Mechanism for Inequality: A Review of Theoretical and Empirical Developments". In: *Annual Review of Sociology* 32.1, pp. 271–297. DOI: 10.1146/annurev.soc.32.061604.123127.

Dowd, Timothy J. (2004). "Concentration and Diversity Revisited: Production Logics and the U.S. Mainstream Recording Market, 1940–1990". In: *Social Forces* 82.4, pp. 1411–1455. DOI: 10.1353/sof.2004.0067.

Ewens, W. J. (1972). "The Sampling Theory of Selectively Neutral Alleles". In: *Theoretical Population Biology* 3.1, pp. 87–112. DOI: 10.1016/0040-5809(72)90035-4.

Fraiberger, Samuel P. et al. (2018). "Quantifying Reputation and Success in Art". In: *Science* 362.6416, pp. 825–829. DOI: 10.1126/science.aau7224.

Guillory, John (1993). *Cultural Capital: The Problem of Literary Canon Formation*. Chicago: University of Chicago Press.

Hannan, Michael T. and John Freeman (1984). "Structural Inertia and Organizational Change". In: *American Sociological Review* 49.2, pp. 149–164. DOI: 10.2307/2095567.

Hoppe, Fred M. (1984). "Pólya-Like Urns and the Ewens' Sampling Formula". In: *Journal of Mathematical Biology* 20.1, pp. 91–94. DOI: 10.1007/BF00275863.

Hubbell, Stephen P. (2001). *The Unified Neutral Theory of Biodiversity and Biogeography*. Monographs in Population Biology 32. Princeton, NJ: Princeton University Press.

Janosov, Milán, Federico Battiston, and Roberta Sinatra (2020). "Success and Luck in Creative Careers". In: *EPJ Data Science* 9.1, 9. DOI: 10.1140/epjds/s13688-020-00227-w.

Kandler, Anne and Enrico R. Crema (2019). "Analysing Cultural Frequency Data: Neutral Theory and Beyond". In: *Handbook of Evolutionary Research in Archaeology*. Ed. by Anna Marie Prentiss. Cham: Springer, pp. 83–108. DOI: 10.1007/978-3-030-11117-5_5.

Leroi, Armand M. et al. (2020). "Neutral Syndrome". In: *Nature Human Behaviour* 4.8, pp. 780–790. DOI: 10.1038/s41562-020-0844-7.

Lopes, Paul D. (1992). "Innovation and Diversity in the Popular Music Industry, 1969 to 1990". In: *American Sociological Review* 57.1, pp. 56–71. DOI: 10.2307/2096144.

Maslinsky, Kirill (2022). *Bibliografiia detskoi knigi [Bibliography of Children's Books] 1918–1984*. Version V5. Repository for Open Data on Russian Literature and Folklore. DOI: 10.31860/openlit-2022.12-B010.

— (Nov. 2026). *Replication Data for: Ageing Institutions Ossify a Cultural Field: Evidence from Soviet Book Publishing for Children*. Version 1.0. Zenodo. DOI: 10.5281/zenodo.24039218.

Mastroianni, Adam (May 2, 2022). *Pop Culture Has Become an Oligopoly*. Experimental History (blog). URL: https://www.experimental-history.com/p/pop-culture-has-become-an-oligopoly (visited on 08/14/2026).

Merton, Robert K. (1968). "The Matthew Effect in Science". In: *Science* 159.3810, pp. 56–63. DOI: 10.1126/science.159.3810.56.

Neiman, Fraser D. (1995). "Stylistic Variation in Evolutionary Perspective: Inferences from Decorative Diversity and Interassemblage Distance in Illinois Woodland Ceramic Assemblages". In: *American Antiquity* 60.1, pp. 7–36. DOI: 10.2307/282074.

Peterson, Richard A. and David G. Berger (1975). "Cycles in Symbol Production: The Case of Popular Music". In: *American Sociological Review* 40.2, pp. 158–173. DOI: 10.2307/2094343.

Salganik, Matthew J., Peter Sheridan Dodds, and Duncan J. Watts (2006). "Experimental Study of Inequality and Unpredictability in an Artificial Cultural Market". In: *Science* 311.5762, pp. 854–856. DOI: 10.1126/science.1121066.

Sørensen, Jesper B. and Toby E. Stuart (2000). "Aging, Obsolescence, and Organizational Innovation". In: *Administrative Science Quarterly* 45.1, pp. 81–112. DOI: 10.2307/2666980.